Impact of Context Quality and Noise in RAG for Open Domain Complex QA

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Abstract

Large language models (LLMs) achieve strong performance on question answering, but remain unreliable for open-domain complex QA that requires multi-hop reasoning. Retrieval-Augmented Generation (RAG) attempts to ground answers in external documents, implicitly assuming that providing relevant contexts is sufficient for accurate answering. In practice, however, retrieved contexts are often noisy and may include irrelevant or misleading passages, and the effect of such negative contexts on downstream generation is not yet systematically understood. We conduct a study of context quality in RAG by injecting random negatives and hard negatives, benchmarking against oracle contexts, and evaluating an ADORE-trained retriever. We find that gold-only oracle contexts achieve much higher accuracy than the best mixed-context retrieval setting (34.92 vs. 21.75 EM), revealing a 13.17-point performance gap. What is more, off-topic random negatives are substantially more harmful to multi-hop answer generation than semantically related hard negatives as retrieval depth increases, indicating that context type plays a critical role in downstream QA performance.

1 Introduction

LLMs are increasingly performing well in open-domain question answering tasks. However, in more complex scenarios, they are still likely to provide unreliable answers, especially when the problem requires multi-hop reasoning across multiple pieces of information wei2022chain. This is because LLMs cannot directly access external evidence. RAG addresses this issue cleverly by incorporating external documents into the generation process lewis2020retrieval, gao2024retrieval. It relies on internal model knowledge, but also adjusts its output based on the retrieved context added to the prompt. Therefore, we are concerned with errors caused by imperfect or noisy retrieved evidence, rather than with proposed changes to the

generative architecture itself. While RAG is therefore essential for complex QA, its robustness is constrained by the quality of the retrieved context.

A key difficulty is that real retrieval outputs are rarely *clean*. The top-k results typically contain different content, such as true supporting evidence, irrelevant background, and passages that are topically similar yet non-supporting or misleading. To reason about these failure modes, we distinguish random negatives and hard negatives. While prior work has studied limitations of long-context utilization (positional sensitivity such as *lost in the middle* liu2023lost), analyses of RAG failures often under-specify the semantic structure of retrieval noise. As a result, different sources of failure (missing evidence, off-topic noise, and on-topic distractors) are frequently combined, making it unclear why RAG can fail even when retrieval appears to be confident ji2023survey.

This issue is especially severe in multi-hop QA. The model must identify and integrate a multi-document evidence chain while ignoring competing spans. Small mistakes early in the reasoning trajectory can propagate, making multi-hop generation very fragile. Interestingly, recent empirical work suggests that adding extra documents does not always harm performance and may sometimes yield slightly positive effects cuconasu2024power. This challenges the simplistic assumption that "irrelevant context always hurts", and motivates a more diagnostic approach that separates which noise types matter, when they matter as k grows, and how they interact with end-task answer correctness.

This paper takes a diagnostic perspective: rather than proposing a new RAG architecture, we vary the type and quality of context to isolate their causal influence on answer generation. We investigate the following research questions:

- RQ1: How do negative contexts impact downstream answer generation performance?
- RQ2: Are negative contexts more important for answer generation than related contexts?
- RQ3: Does providing only gold contexts deteriorate performance compared to mixing with other negative or related contexts?

Through controlled experiments on multi-hop question answering, we analyze how context quality, noise type, and retrieval optimization affect the robustness of retrieval-augmented generation.

2 Related Work

2.1 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) enhances Large Language Models (LLMs) by grounding generation in external knowledge, mitigating hallucinations and improving factual accuracy lewis2020retrieval, gao2024retrieval. In a standard RAG setup, a retriever first identifies relevant documents from a corpus given a query, and a generator conditions its answer on both the query and the retrieved context. Recent work has focused on optimizing this pipeline, exploring the impact of context length, position, and noise on generation quality liu2023lost, yoran2024making.

2.2 Dense Retrieval & Contriever

Dense retrieval models encode queries and documents into a shared latent space, enabling semantic matching beyond lexical overlap. We used **Contriever** izacard2021unsupervised, a dual-encoder architecture trained via unsupervised contrastive learning. Unlike sparse methods (e.g., BM25), Contriever encodes semantic relationships but can be sensitive to *hard negatives*, documents that are semantically similar to the query but factually incorrect.

2.3 Hard Negative Mining & ADORE

Standard dense retrievers often struggle to distinguish between true positives and hard negatives. zhan2021optimizing introduced **ADORE** (Augmenting Dense Retrieval with Hard Negatives), a framework that optimizes the retrieval objective by dynamically sampling hard negatives during training. ADORE freezes the document encoder to maintain efficiency while fine-tuning the query encoder to maximize the margin between gold documents and high-ranking distractors.

2.4 Complex Question Answering

Complex QA tasks, such as multi-hop reasoning, require integrating information from multiple distinct documents. Datasets like `2WikiMultiHopQA` ho2020constructing are designed to evaluate this capability. Unlike single-hop QA, performance in multi-hop settings is highly sensitive to the *reasoning chain*; missing a single link or introducing irrelevant noise can disrupt the entire generation process.

3 Methodology

We implement a RAG pipeline to isolate the effect of context quality on generated answers. In our experiments, we systematically vary the retrieved context but keep the generator fixed.

3.1 Retrieval and Noise Injection

We use **Contriever** izacard2021unsupervised as the dense retriever in our pipeline. Queries and documents are encoded into a shared 768-dimensional embedding space and ranked by cosine similarity using Maximum Inner Product Search. For each query, we retrieve the top-k documents with $k \in \{1, 3, 5\}$.

In mixed-context settings, we scale the amount of noise proportionally to the retrieval depth, using a k *relevant* + k *noise* configuration that yields $2k$ documents per query, of which exactly 50% are negative (noise) documents.

The specific context configurations used to answer each research question (clean, random negatives, hard negatives, and oracle settings) are defined in detail in the Experimental Setup (Section 4).

3.2 ADORE Fine-Tuning

To check whether improved dense retrieval translates into better RAG performance, we fine-tune the Contriever query encoder using the **ADORE** framework zhan2021optimizing. The document encoder is frozen, and all 526.329 document embeddings are precomputed once, so only the query encoder is updated.

For each training query q, we treat the gold supporting documents as positives d^+ and sample hard negatives d^- from the top-N retrieved documents under the current query encoder, excluding all positives and trivially easy candidates. The document encoder is kept fixed, and only the query encoder is fine-tuned.

We optimize a loss:

$$\mathcal{L}(d^+, d^-) = M \cdot \log\left(1 + e^{\left(f(q,d^-) - f(q,d^+)\right)}\right), \tag{1}$$

where $f(q, d)$ denotes the cosine similarity between the query and document embeddings.

The weighting term

$$M = \left|\frac{1}{\mathrm{rank}(d^-)} - \frac{1}{\mathrm{rank}(d^+)}\right| \tag{2}$$

upweights training pairs whose relative ordering has a larger impact on the final ranking, emphasizing errors involving highly ranked negatives or lower-ranked

positives. In practice, we apply a small positive floor (0.05) to M to stabilize training.

Optimization is performed using AdamW with a learning rate of 1×10^{-5}, a batch size of 64, and training for 3 epochs over 167454 queries. After fine-tuning, top-k documents retrieved by the trained ADORE query encoder are passed unchanged to the generator.

3.3 Generator Model

We use **Flan-T5-Base** chung2022scaling, an instruction-tuned encoder–decoder model, as a fixed generator across all experiments. For each query, the model receives a single input sequence consisting of an instruction, the retrieved documents, and the question. Retrieved documents are included and formatted as separate context blocks (e.g., *Context 1*, *Context 2*, etc.). The instruction explicitly asks the model to answer in the shortest form possible and to rely only on the provided context. During inference, we use greedy decoding with a maximum output length of 50 tokens.

The prompt is of the form:

> *Answer the question in the shortest form possible (a few words). Use only the context provided.*
>
> **Context 1:** ...
> **Context 2:** ...
>
> **Question:** q

4 Experiments

Our experiments are designed to answer the three research questions introduced in Section 1. We keep the generator fixed and systematically vary the retrieved context to isolate the effect of context quality and noise on downstream answer generation. To answer RQ1 and RQ2, we inject different types of negative contexts (random and hard negatives) into the retrieved evidence and analyze their effect on downstream answer generation. To answer RQ3, we compare retriever-based RAG systems against an oracle setting with gold-only contexts, and further check whether improving retrieval quality through ADORE fine-tuning can reduce the gap to this upper bound.

4.1 Dataset

We conduct experiments on the **2WikiMultiHopQA** dataset ho2020constructing, which contains open-domain questions requiring multi-hop reasoning across multi-

ple Wikipedia articles. We evaluate on the first 1200 examples from the development set. The underlying Wikipedia corpus consists of 526329 documents.

To avoid treating semantically identical documents as different contexts, we normalize document titles by lowercasing and removing punctuation, and remove documents with identical normalized content. This preprocessing step eliminates approximately 6% of the corpus and is particularly important for hard negative construction and ADORE training.

4.2 Context Configurations

For each query, we retrieve the top-k documents using Contriever, with $k \in \{1, 3, 5\}$. We then construct different context configurations to answer the research questions:

- **Clean Baseline:** The generator receives only the top-k documents retrieved by Contriever.
- **Random Negatives:** To study the effect of unrelated distraction, we inject k randomly sampled documents from the corpus, excluding retrieved documents. This yields a k relevant $+ k$ random noise setting.
- **Hard Negatives:** To study the effect of semantically related but non-supporting evidence, we inject k high-ranking documents beyond the top-k whose titles do not match any gold supporting fact. This yields a k relevant $+ k$ hard negative setting.
- **ADORE Retrieval:** We replace the Contriever query encoder with an ADORE-fine-tuned encoder and retrieve top-k documents, keeping the generator and context formatting unchanged. It tests whether improved retrieval ranking reduces the gap to oracle performance.

In all noisy settings, the total context size is fixed at $2k$ documents, making sure that observed effects are attributable to context quality, not context length.

4.3 Oracle Upper Bound

To quantify the upper bound imposed by context quality, we evaluate an **oracle** setting in which only the annotated 'gold' supporting documents are provided as context. This setting removes retrieval noise entirely and represents the best achievable performance given perfect evidence selection.

4.4 Evaluation Metrics

We evaluate downstream answer generation using standard open-domain QA metrics rajpurkar2016squad:

- **Exact Match (EM):** The percentage of predictions that exactly match the gold answer after normalization.
- **Token-level F1:** The average token overlap between predicted and gold answers.

4.5 Hardware Details

All experiments are conducted on an NVIDIA RTX 4090 GPU with 24GB of VRAM using `vast.ai` cloud computing platform. All results are obtained from a single deterministic run.

5 Results

In this section, we present the experimental results of our RAG pipeline on the 2WikiMultiHopQA dataset. We evaluate the performance using Exact Match (EM) and F1 Score. Our analysis focuses on comparing the baseline retrieval performance against an Oracle upper bound and investigating the impact of different types of noise (Random vs. Hard Negatives) on the generation quality.

5.1 Impact of negatives (random vs. hard)

Q1 "How do negative contexts impact downstream answer generation performance?"

We evaluated the model's robustness by injecting random noise and hard negatives. We scaled the noise proportionally with retrieval depth (k relevant + k noise).

Random Noise Causes Semantic Collapse. As hypothesized, random noise is detrimental. The EM scores drop slightly at TOP 1 and 3, but impact becomes catastrophic at context saturation ($k = 5$), where the Exact Match score plummets by 4.25 percentage points (from 21.75% to 17.50%). This confirms that semantically irrelevant information causes severe semantic fragmentation, breaking the reasoning chain and distracting the generator with unrelated topics. This indicates that the existence of a context saturation mechanism, where adding more paragraphs does not generate more useful evidence, but instead exacerbates noise. Therefore,

Table 1: **Oracle vs. Mixed-Context RAG Performance on 2WikiMultiHopQA.** Exact Match (EM) and F1 scores for gold-only oracle contexts and mixed-context retrieval settings at different retrieval depths. ΔEM denotes the change relative to the Contriever baseline at the same depth.

Context Setting	System / Variation	Depth	EM (%)	F1 (%)	Δ EM
Gold Only	**Oracle (Upper Bound)**	-	**34.92**	**44.28**	*Ceiling*
Mixed Contexts	Baseline (Contriever)	Top-1	21.00	26.07	-
	+ ADORE (Optimized)		**21.67**	**27.22**	+0.67
	+ Random Noise (1+1)		20.42	24.61	-0.58
	+ Hard Negative (1+1)		20.42	25.19	-0.58
Mixed Contexts	Baseline (Contriever)	Top-3	21.58	26.36	-
	+ ADORE (Optimized)		20.25	25.32	-1.33
	+ Random Noise (3+3)		21.17	25.58	-0.41
	+ Hard Negative (3+3)		**21.75**	**26.86**	+0.17
Mixed Contexts	Baseline (Contriever)	Top-5	**21.75**	**27.00**	-
	+ ADORE (Optimized)		15.00	19.37	-6.75
	+ Random Noise (5+5)		17.50	20.55	-4.25
	+ Hard Negative (5+5)		21.33	25.95	-0.42

semantically irrelevant information leads to semantic fragmentation and distracts the generator's attention with unrelated topics.

Hard Negatives are Surprisingly Stable. In contrast, Hard Negatives have a negligible or even positive impact. At Top-5, while Random Noise caused a 4.25% drop, Hard Negatives only reduced performance by 0.42%. Most surprisingly, at Top-3, injecting Hard Negatives actually improved the baseline performance (+0.17%). This may be because hard negatives have a topic-preserving function. They remain semantically close to the query, maintaining topic coherence rather than introducing abrupt topic shifts.

Answering RQ1. Generation can be viewed as a mixture of retrieved documents, with the mixing weights determined by the retrieval system. Therefore, injecting negative words will disrupt this document mixing and change the distribution of the generated documents. The impact of negative contexts is highly dependent on semantic relevance. Negative contexts do not uniformly degrade performance: random noise acts as a severe distractor, especially at higher context lengths, verifying that the model struggles to filter explicitly irrelevant information. Hard negatives

are much less harmful. This implies that the generator's primary bottleneck is not the mere presence of non-gold documents, but rather the loss of semantic coherence caused by random topic shifts.

5.2 RQ2: Are negative contexts more important for answer generation than related contexts?

Random Noise is More Detrimental. Our results indicate that unrelated random noise is far more harmful to generation performance than semantically related hard negatives. This trend is most visible at context saturation ($k = 5$): while adding Random Noise caused a significant performance collapse (ΔEM $= -4.25\%$), adding Hard Negatives resulted in a negligible decrease (ΔEM $= -0.42\%$). This notable difference in degradation demonstrates that the generator is highly resilient to "related" noise but extremely fragile to "unrelated" distraction.

Hard Negatives and Semantic Stability. Contrary to the intuition that hard negatives act as "deceptive traps" that lure the model into incorrect answers, we find they are relatively harmless. In the Top-3 setting, hard negatives even slightly improved performance over the clean baseline (+0.17% EM). We hypothesize that because hard negatives share high semantic overlap with the query (being selected by the dense retriever), they act as "soft context," reinforcing the topical field of the query without actively disrupting the reasoning flow.

Our findings contrast with recent works suggesting that random noise can improve RAG performance by preventing attention collapse. We attribute this discrepancy to the specific demands of *multi-hop* reasoning. Unlike single-hop tasks, multi-hop QA requires the model to bridge information across distinct documents Ho2020. Random Noise introduces *Semantic Fragmentation*: physically separating related evidence pieces in the context window with off-topic text. Hard negatives, by intrinsically being *on-topic*, preserve this semantic coherence, allowing the model to filter facts more effectively than it can filter severe topic shifts.

Answering RQ2. Yes, unrelated contexts are significantly more harmful. Random negatives induce substantially larger degradation in downstream answer generation than related contexts, particularly at larger retrieval depths. We conclude that *semantic coherence* is a critical factor for generation robustness; as long as the context remains topically consistent, the generator can tolerate factual distractions, but it struggles significantly when faced with the semantic discontinuity introduced by random noise.

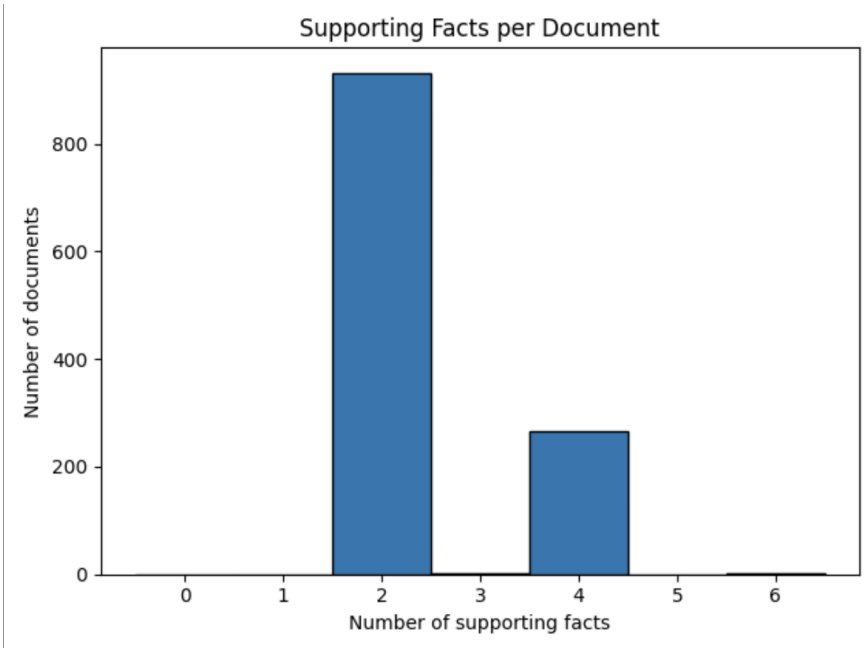

Figure 1: **Distribution of supporting facts per top 1200 documents in 2WikiMultiHopQA.** Most documents contain two supporting facts, indicating that multi-hop evidence is often densely concentrated within a small number of documents.

5.3 Gold-only vs. retrieved contexts

RQ3: Does providing only gold contexts deteriorate performance compared to mixing with other contexts?

To answer RQ3, we first establish an upper bound on downstream QA performance using an oracle setting. In this configuration, the generator is provided only with the annotated gold supporting documents for each question. These oracle documents are guaranteed to contain the evidence required to answer the question and, therefore, represent the best possible context selection scenario. As expected, the oracle setting achieves strong performance, confirming that the generator is capable of answering questions accurately when supplied with complete and noise-free evidence.

We then compare this oracle performance to a retrieval-based baseline using Contriever. In this setting, the retrieved context is inherently mixed: depending on the query, some retrieved documents are highly relevant, while others are only partially relevant or irrelevant. Compared to the oracle, the Contriever-based system performs substantially worse, indicating that retrieval noise and missing evidence

significantly degrade downstream answer generation.

To investigate whether this gap could be reduced through improved retrieval quality, we evaluate an ad-hoc retriever trained using the ADORE framework. The motivation behind this experiment is to test whether a retriever optimized to better distinguish relevant documents from hard negatives can narrow the performance gap to the oracle. Empirically, we observe that ADORE improves performance at shallow retrieval depth (Top-1), suggesting that better ranking of the single most relevant document can benefit downstream QA. However, this improvement does not extend to larger retrieval depths. At Top-3 and Top-5, performance decreases relative to the Contriever baseline.

To better understand this behavior, we conduct a post-hoc analysis of the oracle evidence structure by examining how many supporting facts are contained within each gold document. Figure 1 shows the distribution of supporting facts per document for the top 1200 examples. We observe that, for most questions, the required evidence is concentrated in a small number of documents, with the majority of oracle documents containing two supporting facts. This indicates that multi-hop evidence is often densely packed rather than distributed across many documents.

Based on this observation, we hypothesize that retrieving a larger number of documents, particularly with a retriever optimized for ranking confidence, such as ADORE, can introduce highly plausible but incomplete or misleading contexts. When the generator receives five retrieved documents for a query that effectively requires only a small number of dense evidence documents, the additional contexts may compete with or mask the true supporting information. This effect is amplified when the retriever confidently ranks semantically strong but non-supporting documents, which can mislead the generator and degrade performance at higher retrieval depths.

Answering RQ3. Providing only gold contexts does not deteriorate performance. On the contrary, gold-only oracle contexts yield substantially higher accuracy than any mixed-context retrieval setting. The observed performance gap suggests that mixing retrieved documents with non-gold contexts introduces interference rather than useful additional evidence, particularly when the number of retrieved documents exceeds the number of documents that actually contain supporting facts.

5.4 Qualitative Analysis: Case Study

To better understand how hard negatives can mislead the model, we analyze a representative example where the baseline system answers correctly, but the model augmented with hard negatives fails.

Case Study: Semantic Distraction. Consider the following comparison question from the development set (ID: `27987581093611ebbdaeac1f6bf848b6`):

> **Question:** "Which film came out earlier, *Niyoti* or *Angali Pangali*?"
> **Gold Answer:** Niyoti
> **Baseline Prediction (Correct):** Niyoti
> **With Hard Negatives (Incorrect):** Angali Pangali

Analysis of Failure. In the clean baseline setting, the retrieved context contains sufficient evidence to compare release dates, allowing the model to produce the correct answer. After injecting hard negatives, however, the context includes high-plausibility distractor documents that mention the competing entity (*Angali Pangali*). These distractors are semantically related to the query and therefore cannot be dismissed as irrelevant.

As a result, the model is presented with multiple plausible answer candidates supported by different parts of the context. Since our generator uses greedy decoding, generation proceeds along a single, locally optimal trajectory. Once early tokens align with a distractor-consistent hypothesis, the decoder cannot revise this decision in later steps, even if gold evidence is present elsewhere in the context. Consequently, the model commits to the distractor-supported answer and outputs *Angali Pangali*.

This example shows how hard negatives can induce errors by introducing semantically plausible competing evidence, which interferes with evidence aggregation during generation.

6 Discussion

Our results show that the effectiveness of Retrieval-Augmented Generation for complex QA is governed by retrieval accuracy, and more fundamentally by the semantic structure of the retrieved context. Across all three research questions, we observe that context quality, coherence, and evidence density jointly determine downstream answer generation performance.

Context noise is harmful, but not uniformly so. Addressing RQ1 and RQ2, we find that the impact of negative contexts depends critically on their semantic relationship to the query. Random negatives, which are semantically unrelated, consistently degrade performance and cause severe failures at higher retrieval depths. This suggests that multi-hop reasoning is particularly vulnerable to semantic fragmentation: when irrelevant documents interrupt the topical flow of evidence, the generator struggles to maintain a coherent reasoning chain. In contrast, hard negatives—while

factually incorrect - remain topically aligned with the query and are therefore far less disruptive. In some settings, they even yield slight improvements, likely by reinforcing the semantic field without introducing abrupt topic shifts.

Oracle performance - a structural upper bound. For RQ3, the oracle setting establishes a performance ceiling, demonstrating that the generator is capable of strong multi-hop reasoning when provided with clear, complete evidence. However, this performance is not approached by any retriever-based configuration. A key insight emerges from our post-hoc analysis of oracle documents (Figure 1): most questions require evidence that is densely concentrated within a small number of documents, typically containing two supporting facts. This implies that oracle contexts differ structurally from retrieved contexts, not merely in correctness but in evidence density.

Why deeper retrieval can hurt. This structural mismatch helps explain why increasing retrieval depth does not close the oracle gap and why ADORE degrades performance at Top-3 and Top-5. When a question effectively requires only a small number of dense evidence documents, retrieving additional documents introduces competition rather than complementary information. Retrievers optimized for ranking confidence may surface semantically strong but incomplete documents that distract the generator. As context size grows, the generator must arbitrate among more plausible yet conflicting evidence fragments, increasing the risk of committing to an incorrect reasoning path.

Retrieval-generation misalignment. Taken together, these findings point to a misalignment between standard retrieval objectives and generation utility in multi-hop QA. Improvements in retrieval ranking precision, such as those achieved by ADORE at shallow depth, do not necessarily translate into better downstream performance when multiple documents are required. This suggests that retrieval should be evaluated not only on relevance or rank-based metrics, but also on its ability to supply a coherent and sufficient evidence set for generation.

Implications. Our study indicates that effective RAG for complex QA requires more than retrieving *better* documents: it requires retrieving the *right number* of documents with the *right evidence structure*. Future RAG systems may benefit from generator-aware retrieval, evidence aggregation objectives, or adaptive context selection mechanisms that account for evidence density and redundancy, rather than blindly increasing retrieval depth.

Future Work Our findings highlight a critical misalignment between retrieval metrics and generation utility, where optimizing for strict ranking precision (ADORE) degraded downstream performance at larger context depths (k=5). Future work should address this by exploring **end-to-end training objectives**, where the retriever is optimized directly for answer correctness or evidence coverage rather than proxy ranking losses. Additionally, given the generator's significant sensitivity to random noise, we propose investigating **noise-aware fine-tuning** for the generator or scaling to larger architectures (e.g., Llama-3, Mistral) to determine if robust context filtering is an emergent capability that can better stabilize performance in high-noise open-domain settings. Finally, a more realistic upper bound could be established by evaluating hybrid oracle settings that mix gold documents with controlled noise, allowing a finer analysis of how much non-gold context can be tolerated before generation degrades.

7 Conclusion

In this report, we analyzed the impact of context quality on RAG systems for complex QA. Our experiments on the 2WikiMultiHopQA dataset yielded three key insights:

1. **Random Noise:** Contrary to some recent findings, random noise consistently degrades performance for our Flan-T5 model, highlighting the importance of noise filtering.

2. **Hard Negatives:** Surprisingly, hard negatives are less harmful than random noise in high-context settings, suggesting that maintaining semantic coherence is more critical than avoiding all distractions.

3. **Retrieval Gap:** A significant gap remains between retriever-based systems and the Oracle upper bound, underscoring retrieval as the primary bottleneck.

A Contributions

Table 2 summarizes the task-based contributions of each author to this work.

Table 2: Who did what: task-based author contributions

Task	**Amelia**	**Claire**	**Samuel**
Data preprocessing	✓		✓
Contreiver-based data extraction	✓	✓	
Answer generation and evaluation metrics		✓	✓
Oracle experiment design and execution	✓		✓
Experiments with random negatives and hard negatives		✓	✓
Contriever fine-tuning with ADORE	✓		✓
Report Writing	✓	✓	✓